AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Basic training and test accuracy



Session 8 of 10

Outline

- Evaluating forecast accuracy
- Evaluating point forecast accuracy
- Evaluating distributional forecast accuracy

SECTION 1 OF 3

Evaluating forecast accuracy

1. **Evaluating forecast accuracy**
2. Evaluating point forecast accuracy
3. Evaluating distributional forecast accuracy



Evaluate forecast accuracy

- In order to evaluate the performance of a forecasting model, we compute its forecast accuracy.
- Forecast accuracy is compared by measuring errors based on the test set.
- Ideally it should allow comparing benefits from improved accuracy with the cost of obtaining the improvement.

Evaluate forecast accuracy- Business impact

- We should be choosing forecast models that lead to better business decisions
  - least staffing costs, least emission, highest service level, least stock-out, least inventory, fastest response, least change in planing, for example.
- However, this is not always easy to obtain, therefore we might simply use methods that provide the most accurate forecast.

In-sample (training) vs. out-of-sample (test)

- Fitting and its residual are not a reliable indication of forecast accuracy
- A model which fits the training data well will not necessarily forecast well
- A perfect fit can always be obtained by using a model with enough parameters
- Over-fitting a model to data is just as bad as failing to identify a systematic pattern in the data

Forecast accuracy evaluation using test sets

- We mimic the real life situation
- We pretend we don’t know some part of data (new data)
- It must not be used for *any* aspect of model training
- Forecast accuracy is computed only based on the test set

Training and test sets



SECTION 2 OF 3

Evaluating point forecast accuracy

1 Evaluating forecast accuracy
2 **Evaluating point forecast accuracy**
3 Evaluating distributional forecast accuracy



Forecast errors

Forecast “error”: the difference between an observed value and its forecast.

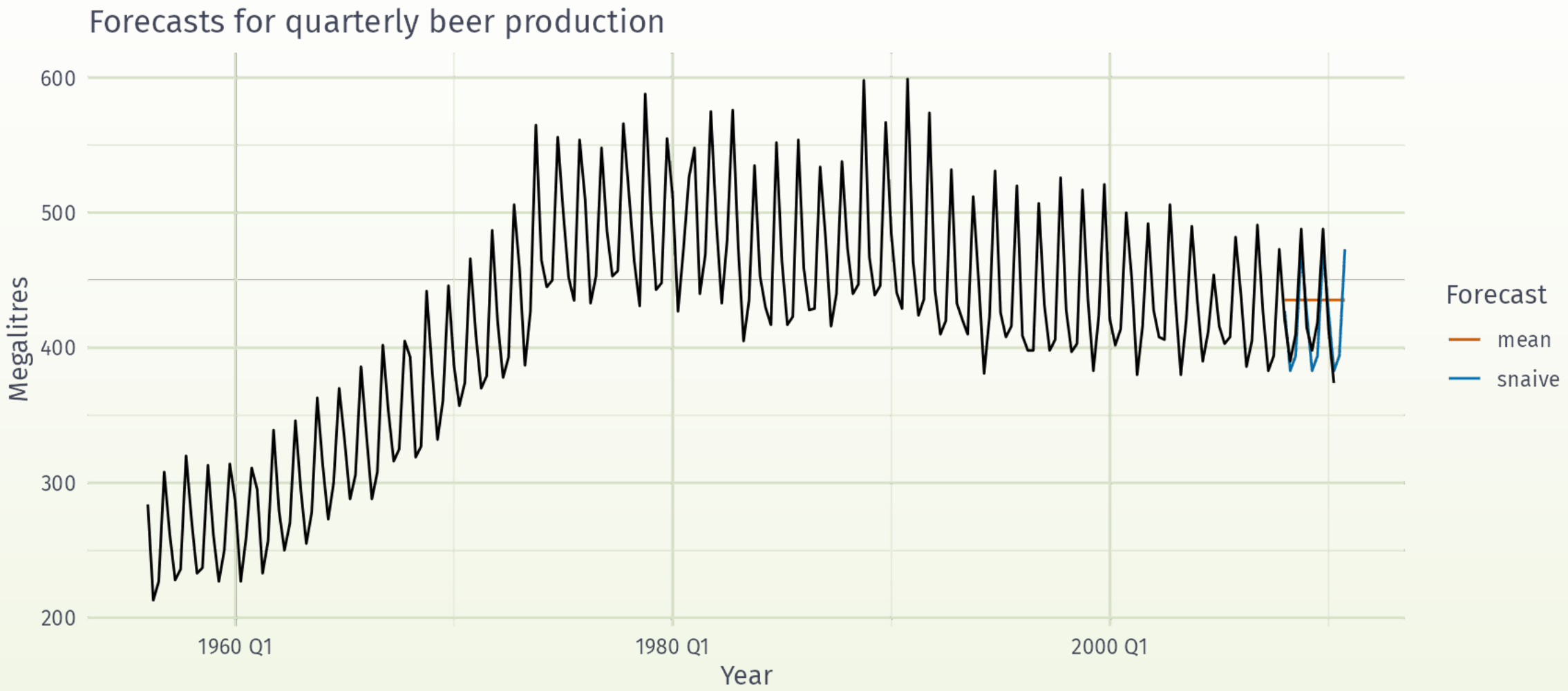
$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T},$$

where the training data is given by $\{y_1, \dots, y_T\}$

Measures of forecast accuracy

```
beer_fit <- aus_production |>
  filter(between(year(Quarter), 1992, 2007)) |>
  model(
    snaive = SNAIVE(Beer),
    mean = MEAN(Beer)
  )
beer_fit |>
  forecast(h = "3 years") |>
  autoplot(aus_production, level = NULL) +
  labs(title ="Forecasts for quarterly beer production",
       x ="Year", y ="Megalitres") +
  guides(colour = guide_legend(title = "Forecast"))
```

Measures of forecast accuracy



Measures of forecast accuracy

$$y_{T+h} = (T+h)\text{th observation}, h = 1, \dots, H$$
$$\hat{y}_{T+h|T} = \text{its forecast based on data up to time } T.$$
$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

$$\text{MAE} = \text{mean}(|e_{T+h}|)$$
$$\text{MSE} = \text{mean}(e_{T+h}^2) \qquad \text{RMSE} = \sqrt{\text{mean}(e_{T+h}^2)}$$
$$\text{MAPE} = 100\text{mean}(|e_{T+h}|/|y_{T+h}|)$$

- MAE, MSE, RMSE are all scale dependent.
- MAPE is scale independent but is only sensible if $y_t \gg 0$ for all t, and y has a natural zero.

Measures of forecast accuracy

Mean Absolute Scaled Error

$$\text{MASE} = \text{mean}(|e_{T+h}|/Q)$$

- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1}\sum_{t=2}^{T}|y_t - y_{t-1}|$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m}\sum_{t=m+1}^{T}|y_t - y_{t-m}|$$

where m is the seasonal frequency

Proposed by Hyndman and Koehler (IJF, 2006).

Measures of forecast accuracy

Root Mean Squared Scaled Error

$$\text{RMSSE} = \sqrt{\text{mean}(e_{T+h}^2/Q)}$$

- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1}\sum_{t=2}^{T}(y_t - y_{t-1})^2$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m}\sum_{t=m+1}^{T}(y_t - y_{t-m})^2$$

where m is the seasonal frequency

Proposed by Hyndman and Koehler (IJF, 2006).

Measures of forecast accuracy

```
beer_fc <- forecast(beer_fit, h = "3 years")
accuracy(beer_fc, aus_production)
```

```
# A tibble: 2 × 10
  .model .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE    ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl>
1 mean   Test  -13.8  38.4  34.8 -3.97  8.28 2.20  1.96  -0.0691
2 snaive Test    5.2  14.3  13.4  1.15  3.17 0.847 0.729  0.132
```

SECTION 3 OF 3

Evaluating distributional forecast accuracy

1 Evaluating forecast accuracy
2 Evaluating point forecast accuracy
3 **Evaluating distributional forecast accuracy**



Prediction interval accuracy using winkler score

Winkler proposed a scoring method to enable comparisons between prediction intervals:

- it takes account of both coverage and width of the intervals.

Winkler score

$$W(l_t, u_t, y_t) = \begin{cases} u_t - l_t & \text{if } l_t < y_t < u_t \\ (u_t - l_t) + \frac{2}{\alpha}(l_t - y_t) & \text{if } y_t < l_t \\ (u_t - l_t) + \frac{2}{\alpha}(y_t - u_t) & \text{if } y_t > u_t \end{cases}$$

Prediction interval accuracy

```
  # Compute interval accuracy
beer_fc |> accuracy(aus_production,
   measures = interval_accuracy_measures)
```

```
# A tibble: 2 × 5
  .model .type winkler pinball scaled_pinball
  <chr>  <chr>   <dbl>   <dbl>          <dbl>
1 mean   Test    174.     8.67         0.274
2 snaive Test     86.3    3.10         0.0980
```

Quantile score

Quantile score

$$Q_{p,t} = \begin{cases} 2(1-p)\big(f_{p,t} - y_t\big), & \text{if } y_t < f_{p,t} \\ 2p\big(y_t - f_{p,t}\big), & \text{if } y_t \geq f_{p,t} \end{cases}$$

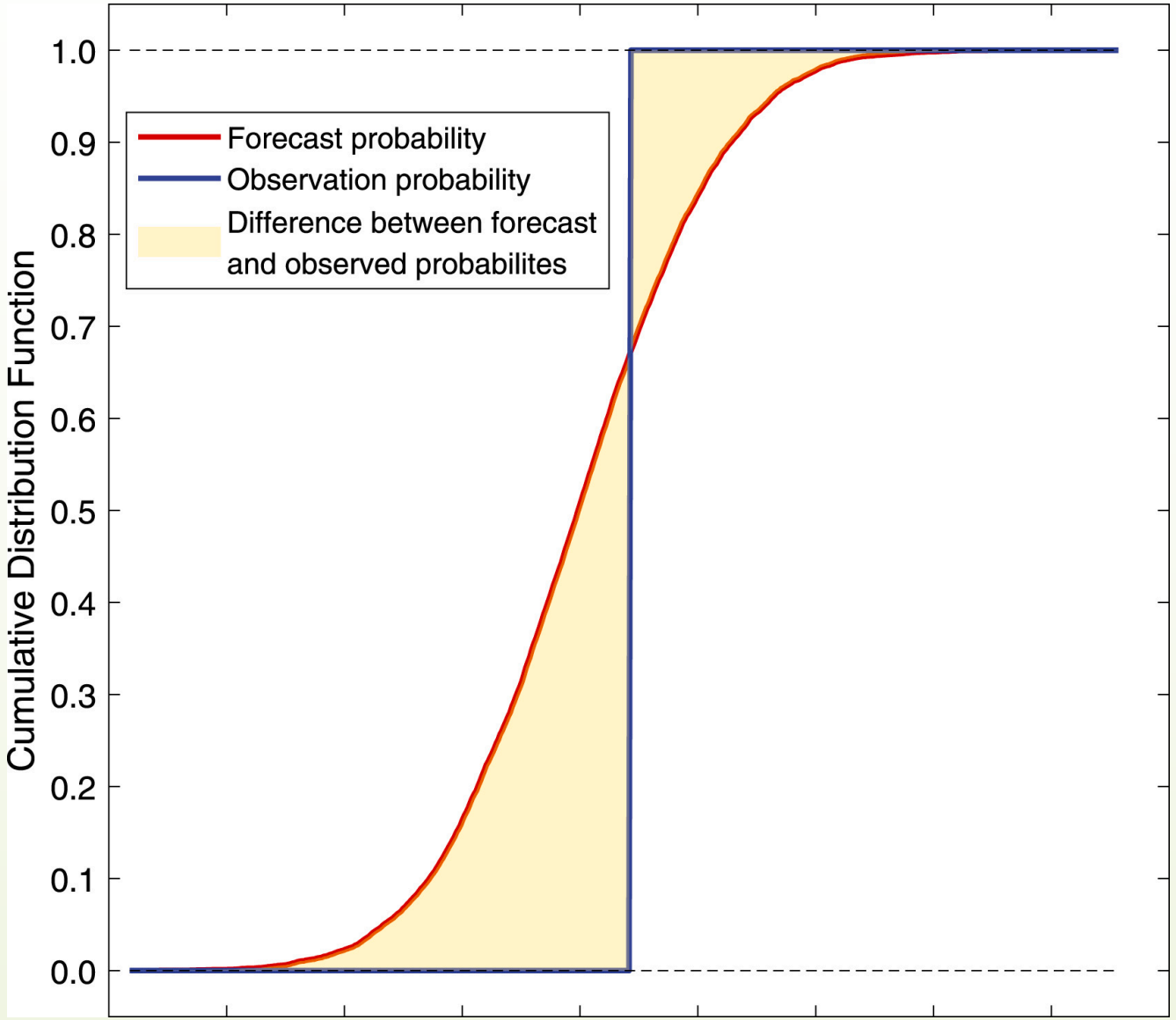
Continuous Ranked Probability Score (CRPS)

$$\mathrm{CRPS} = \mathrm{mean}(p_j),$$

where

$$p_j = \int_{-\infty}^{\infty} \left(G_j(x) - F_j(x)\right)^2 dx,$$

Continuous Ranked Probability Score (CRPS)



Quantile score and CRPS

```
beer_fc |>
  accuracy(aus_production, list(measures=distribution_accuracy_measures))
```

```
# A tibble: 2 × 4
  .model .type measures.percentile measures.CRPS
  <chr>  <chr>                <dbl>         <dbl>
1 mean   Test                 22.7          22.4
2 snaive Test                  8.80          8.71
```

Quantile score

```
beer_fc |>
  accuracy(aus_production, list(qs=quantile_score), probs = .9)
```

```
# A tibble: 2 × 3
  .model .type    qs
  <chr>  <chr> <dbl>
1 mean   Test  14.1
2 snaive Test   4.60
```